

Discrete Assignment - 3

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Q1ai) If $a|b$ and $a|c$ then $a|(b+c)$

Sol

$$b = as \quad \text{and} \quad c = at$$

Then

$$b+c = as+at = a(s+t)$$

Since $s+t$ is an integer, then a divides $(b+c)$.

$$a|(b+c) \quad \text{proved}$$

ii) If $a|b$ then $a|(bc)$

Sol

$$b = ak \quad (\text{for some integer "k"})$$

multiply both side by "c"

$$bc = (ak)c$$

$$bc = a(kc)$$

Since kc is an integer, then a divides (bc) .

$$a|(bc) \quad \text{proved}$$

iii) If $a|b$ and $b|c$, then $a|c$

Sol

$$b = au \quad \text{and} \quad c = bv$$

Substitute

$$c = (au)v$$

$$c = a(uv)$$

Since uv is an integer, then a divides c

$$a|c \quad \text{proved.}$$

Q1b) Product of two consecutive integer is even

Sol:- Let 'n' be an integer

$n, (n+1)$ are consecutive integer

Let $n = 2k$ for some integer k . n is even

$\therefore n+1 = 2k+1$, this will be odd consecutive value

so substitute

$$n(n+1) \Rightarrow \cancel{2k(k+1)} \quad 2k(2k+1)$$

$$\cancel{2(k(k+1))}, \quad 2(k(2k+1)) \text{ where}$$

this is divisible by 2, hence even.

Therefore the product of two consecutive integers is even.

Since $k(2k+1)$ is an integer so it can be denoted as "r".

Therefore the answer is $2r$ which is the definition of an even number.

Hence proved that the product of two consecutive integers is even.

Case 2: first number odd, second number even

$$n = 2k+1$$

$$n(n+1) \Rightarrow (2k+1)(2k+2)$$

$$(n+1) = 2k+1+1 = 2k+2$$

$$2((2k+1)(k+1))$$

Since $(2k+1)(k+1)$ is an integer, the product is in the form $2r$ which is also an even number.

Therefore in both cases product of two consecutive integers is even.

Q1c) Product of two odd integers and one even integer is even

$$\begin{array}{ll} \text{Let } x = 2n+1 & \left. \begin{array}{l} y = 2s+1 \\ z = 2t \end{array} \right\} \begin{array}{l} \text{odd number} \\ \text{even number} \end{array} \end{array} \quad \text{where } n, s, t \text{ are integers}$$

$$\begin{aligned} xyz &= (2n+1)(2s+1)(2t) \\ &= 2((2n+1)(2s+1)t) \end{aligned}$$

Since $(2n+1)(2s+1)t$ is an integer we can denote it as "r" and can write $2r$, which is the definition of an even number.

As the product is divisible by 2. It's an even number.
PROVED

Q2a) $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

sol $(\sqrt{a+b})^2 = (\sqrt{a} + \sqrt{b})^2$

$$a+b = a+b+2\sqrt{a}\sqrt{b}$$

$$0 = 2\sqrt{ab}$$

$$(0)^2 = (\sqrt{ab})^2$$

$$0 = ab$$

So, at least one of "a" or "b" must be 0 and other to be greater or equal to 0. To get real solution.

eg $(a=0, b=4)$ $(a=9, b=0)$ $(a=0, b=0)$. In these equality hold.

Q2b) Let r be an arbitrary rational number.

Let $a, b \in \mathbb{Z}$ where $b \neq 0$

sol:

$$r = \frac{a}{b}$$

$$r^2 = \frac{a^2}{b^2}$$

since a^2 and b^2 are integers and $b^2 \neq 0$ so r^2 is a ratio of integers that is rational. PROVED

Q2c) Let $n=7$

$$2^7 - 1 = 128 - 1 = 127$$

check if 127 is prime. $\sqrt{127} = 11.26$ so divide 127 by primes till 11 which is 2, 3, 5, 7, 11

$$127 \div 2 = 63.5 \quad \text{remainder } 1$$

$$127 \div 3 = 42.33 \quad \text{not integer remainder } 1$$

$$127 \div 5 = 25 \quad \text{remainder } 2$$

$$127 \div 11 = 11.54 \quad \text{remainder } 6$$

Since no prime divides 127. Therefore 127 is prime.

Hence statement of an $n > 5$ being a prime is true

Q3a) p : n and m are odd integers
 q : $n+m$ is an even integer

original: $p \Rightarrow q$

contradiction $p \Rightarrow \neg q$ where $\neg q$: $n+m$ is odd integer

Sol

$n = 2k+1$ where k, l are integers

$m = 2s+1$

$$\text{so } n+m = 2k+1 + 2s+1$$

$$= 2k+2s+2$$

$$= 2(k+s+1)$$

Since $k+s+1$ is an integer we can denote it as r so it's in $2r$ format which is definition of even.

Therefore our assumption of contradiction that $(n+m)$ is odd is False and instead the sum is even. Hence proved by contradiction

b) x : p divides a
 y : p does not divide $(a+1)$

a is integer p is prime

original: $x \Rightarrow y$

contradiction: $x \Rightarrow \neg y$ where $\neg y$: p does divide $(a+1)$

Set: $\frac{\text{eq 1}}{p \mid a} \quad \frac{\text{eq 2}}{p \mid (a+1)} \quad \text{for some } k, s \text{ integer}$

$$a = kp \quad (a+1) = sp$$

$$\text{eq 2} - \text{eq 1} \Rightarrow (a+1) - (a) = sp - kp$$
$$1 = p(s-k)$$

This shows that p divides 1 as $(s-k)$ is an integer.

However no prime can divide 1. Hence our assumption of contradiction that $p \mid (a+1)$ is false and that $p \nmid (a+1)$ is true. Hence proved by contradiction.

Q3c) p : set of prime number is infinite.

contradiction:- set of prime number is NOT infinite

Sol:

Let all prime numbers that exists be included in the list,

$$p_1, p_2, p_3, \dots, p_n$$

Let there be a new number N defined as the product of all the prime numbers. plus 1

$$N = (p_1 \times p_2 \times p_3 \times \dots \times p_n) + 1$$

If we divide N by any of the prime numbers then we get a remainder of 1. Therefore either N is a prime number, or it has a prime factor that wasn't included on the list. This contradicts our statement that all the prime numbers were included on a finite list.

Therefore, there are infinitely many prime numbers.

Q4a) if $|x| > 1$ then $x > 1$ or $x < -1$ for all $x \in \mathbb{R}$

$$p: |x| > 1$$

$$q: x > 1 \text{ or } x < -1$$

$$\neg p: |x| \leq 1$$

$$\neg q: -1 \leq x \leq 1 \quad \text{and = range}$$

so the contrapositive says if $-1 \leq x \leq 1$ then $|x| \leq 1$.

This is immediate from the definition of absolute value cause from any x with $-1 \leq x \leq 1$ satisfies $|x| \leq 1$.

Hence, as our contrapositive is true, the original implication is true-

Q4b) If $(m+n)$ is even, then $(m$ and $n)$ ~~are~~ both even or $(m$ and $n)$ are both odd.

$$p: (m+n) \text{ is even}$$

$$\neg p: (m+n) \text{ is odd}$$

q : (m and n) are both even or (m and n) are both odd

$\neg q$: (m and n) are NOT both even and (m and n) are not both odd.

simply $\neg q$ mean one is even and other is odd.

sol: Assume m is even, n is odd

$$m = 2k \quad \text{for some integer } k \text{ and } s$$

$$n = 2s + 1$$

$$(m+n) = 2k + 2s + 1$$
$$= 2(k+s) + 1$$

since $(k+s)$ is an integer we can denote it as ' r ' so we get

$$(m+n) = 2r + 1 \quad \text{where } 2r+1 \text{ is the definition of odd}$$

This proves the contrapositive statement of $(m+n)$ being odd when $\neg q$. Hence the original statement is true.

Q4c) If ab is divisible by 4, then a and b are positive, even integer.

p : ab is divisible by 4

$\neg p$: ab is NOT divisible by 4

q : a and b are positive even integers

$\neg q$: a ^{and} ~~and~~ b are NOT positive even integers.

sol

assume a is ~~even~~ ^{odd} ~~positive~~, b is ~~odd~~ ^{negative} ~~even~~ ^{odd} number

$$a = 2n+1 \quad b = 2m+1$$

~~$ab = (2n)(2n+1)$~~
 ~~$= 4n^2 + 2n$~~

~~$ab = (2n+1)(2n+1)$~~
 ~~$= 4n^2 + 4n + 1$~~

$ab = (2n+1)(2m+1)$

$$ab = 4mn + 2n + 2m + 1$$

$$ab = 2(2mn + m + n) + 1$$

so as ab is form of $2k+1$ for $k = 2mn + m + n$ i.e ab is odd

An odd integer can't be divisible by 4. Hence contrapositive holds. Therefore original statement also holds.

Q5ai) Counterexample :

$$\text{Let } n = 7$$

so $7+2 = 9$ which is not prime

~~ii)~~ $\therefore$ Proposition is false

aii) Counter example

$$\text{Let } n = 6 \text{ even number}$$

$$\text{so } \frac{6}{4} = \text{remainder is } 2$$

$\therefore$ proposition is false

bi) Counter example.

Let $n = 4$ which is positive integer but not prime.

It has 1, 2, 4 as its multiple

bii) Counter example

$$\text{Let } n = \frac{1}{4} \text{ which is } 0.25$$

$$\left(\frac{1}{4}\right)^2 = \frac{1}{16} \text{ which is } 0.0625$$

$$\frac{1}{16} < \frac{1}{4} \text{ so proposition is false}$$

c) Counter example

$$\text{Let } n = \sqrt{2} \text{ irrational}$$

$$p = \sqrt{8} \text{ irrational}$$

$$np = \sqrt{2} \times \sqrt{8}$$

$$= \sqrt{16}$$

$$= 4$$

as the product of two irrational is rational number

$\therefore$ the proposition false.

(Note:- some irrational number product can give irrational answer but it's not always as shown by the counter example).

Q6a) $p: 6-7\sqrt{2}$ is irrational

contradiction: $6-7\sqrt{2}$ is rational. Then there exist integers p, q with $q \neq 0$ and

Sol:

$$\frac{p}{q} = 6-7\sqrt{2}$$

$$7\sqrt{2} = 6 - \frac{p}{q}$$

$$7\sqrt{2} = \frac{6q-p}{q}$$

$$\sqrt{2} = \frac{6q-p}{7q}$$

Since p, q were integer and $q \neq 0$. Therefore RHS meets the definition of a rational number. So our equation states

$$\sqrt{2} = \text{rational Number.}$$

However, this is contradiction as $\sqrt{2}$ is irrational. Therefore by contradiction it is proven that $6-7\sqrt{2}$ is indeed irrational.

b) $p: \sqrt{2} + \sqrt{3}$ is irrational

contradiction: $\sqrt{2} + \sqrt{3}$ is rational.

Sol:

$$\frac{p}{q} = \sqrt{2} + \sqrt{3}$$

$$r = \frac{p}{q}$$

$$(r)^2 = (\sqrt{2} + \sqrt{3})^2$$

$$r^2 = 2 + 3 + 2\sqrt{6}$$

$$r^2 = 5 + 2\sqrt{6}$$

$$r^2 - 5 = 2\sqrt{6}$$

$$\frac{p^2}{q^2} - 5 = 2\sqrt{6}$$

$$\frac{p^2 - 5q^2}{q^2} = \sqrt{6}$$

~~$$\sqrt{3} = -\sqrt{2} + \frac{p}{q}$$
$$(\sqrt{3})^2 = \left(\frac{p - q\sqrt{2}}{q}\right)^2$$~~

~~$$3 = \frac{p^2 + 2q^2 - 2pq\sqrt{2}}{q^2} = \frac{p^2 + 2q^2}{q^2} - \frac{2pq\sqrt{2}}{q^2}$$~~

~~$$3 = \frac{p^2 + 2q^2 - 2pq\sqrt{2}}{q^2}$$~~

since p, q were integer and $q \neq 0$ and $\frac{p}{q}$ was rational representation. Therefore LHS meets the definition of a rational number. so our equation states

$$\text{rational number} = \sqrt{6}$$

However, this is contradiction as $\sqrt{6}$ is irrational. Therefore, by contradiction it is proven that $\sqrt{2} + \sqrt{3}$ is indeed irrational

$$c) 4(z^2 + z + 1) - 3z^2$$

sol

$$4z^2 + 4z + 4 - 3z^2$$

$$z^2 + 4z + 4$$

$$z^2 + 2z + 2z + 4$$

$$z(z+2) + 2(z+2)$$

$$(z+2)(z+2)$$

$$(z+2)^2$$

as $(z+2)^2$ is a perfect square for every integer z . Hence the expression is always a perfect square.

$$Q7a) P(n) = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Base case

for $n = 1$

$$\text{LHS} :- n^2 = (1)^2 = 1$$

$$\text{RHS} :- \frac{n(n+1)(2n+1)}{6} = \frac{1(1+1)(2+1)}{6} = \frac{6}{6} = 1$$

as $\text{LHS} \equiv \text{RHS}$

$1 \equiv 1$ The base case holds

for some $k > 1$

Now Induction

$$P(k) = 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

$$P(k+1) = 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)(k+1+1)(2k+2+1)}{6}$$

Substitute $P(k)$ in $P(k+1)$

$$\frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

solve both LHS and RHS

$$\frac{(k^2+k)(2k+1)}{6} + k^2+2k+1 = \frac{(k^2+3k+2)(2k+3)}{6}$$

$$\frac{2k^3+3k^2+k+6k^2+12k+6}{6} = \frac{2k^3+9k^2+13k+6}{6}$$

$$\frac{2k^3+9k^2+13k+6}{6} = \frac{2k^3+9k^2+13k+6}{6}$$

As LHS is equal to RHS. Thus $P(k+1)$ holds. By induction, the formula is true for all $k \geq 1$.

Q7b) $P(n) = 1^3 + 2^3 + \dots + n^3 = \frac{1}{4} n^2 (n+1)^2$

Base Case:

for $n=1$

$$\text{LHS } P(1) = 1^3 = 1$$

$$\text{RHS} = \frac{1}{4} (1)^2 (1+1)^2 = \frac{4}{4} = 1$$

as $\text{LHS} = \text{RHS}$

$1 = 1$ The base case holds

Now Induction:

for some $k \geq 1$

$$P(k) = 1^3 + 2^3 + \dots + k^3 = \frac{1}{4} k^2 (k+1)^2$$

$$P(k+1) = 1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \frac{1}{4} (k+1)^2 (k+1+1)^2$$

substitute $P(k)$ in $P(k+1)$

$$\frac{1}{4} k^2 (k+1)^2 + (k+1)^3 = \frac{(k+1)^2 (k+2)^2}{4}$$

solve both LHS and RHS

$$\frac{k^2(k^2+2k+1)}{4} + (k+1)(k^2+2k+1) = \frac{(k^2+2k+1)(k^2+4k+4)}{4}$$

$$(k^2+2k+1) \left[\frac{k^2}{4} + (k+1) \right] = \frac{(k^2+2k+1)(k^2+4k+4)}{4}$$

$$(k^2+2k+1) \left[\frac{k^2+4k+4}{4} \right] = \frac{(k^2+2k+1)(k^2+4k+4)}{4}$$

$$\frac{(k^2+2k+1)(k^2+4k+4)}{4} = \frac{(k^2+2k+1)(k^2+4k+4)}{4}$$

As LHS is equal to RHS, Thus $P(k+1)$ holds. By induction, it holds for all $k \geq 1$.

Q7c) $P(n) = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

Base Case

for $n=1$

$$LHS = \frac{1}{1(1+1)} = \frac{1}{2}$$

$$RHS = \frac{n}{n+1} = \frac{1}{1+1} = \frac{1}{2}$$

as $LHS = RHS$

$1 = 1$ The base case holds

Now Induction

for $k \geq 1$

$$P(k) = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

$$P(k+1) = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+1+1)} = \frac{(k+1)}{(k+1)+1}$$

Substitute $P(k)$ in $P(k+1)$

$$\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

Solve both LHS and RHS

$$\frac{k(k+2) + 1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

$$\frac{k^2 + 2k + 1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

$$\frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

$$\frac{k+1}{k+2} = \frac{k+1}{k+2}$$

As LHS = RHS. This $P(k+1)$ holds. By induction it holds for all $k \geq 1$

Q8(a) Quotient and remainder when 19 is divided by 7

Sol

$$19 = 2(7) + 5$$

$$a = qm + r$$

$q = \text{quotient}$
 $r = \text{remainder}$

$\therefore$ Quotient = 2 Remainder = 5 *Ans*

b) -111 is divided by 11

Sol

$$-111 = -11(11) + 10$$

$\therefore$ Quotient = -11 Remainder = 10 *Ans*

c) 789 divided by 23

Sol

$$789 = 34(23) + 7$$

$\therefore$ Quotient = 34 Remainder = 7 *Ans*

d) 1001 divided by 13

Sol

$$1001 = 77(13) + 0$$

$\therefore$ Quotient = 77 Remainder = 0 *Ans*

e) 10 divided by 19

Sol

$$10 = 0(19) + 10$$

$\therefore$ Quotient = 0 Remainder = 10 *Ans*

8f) 3 divided by 5

sol

$$3 = 0(5) + 3$$

$\therefore$ Quotient = 0 Remainder = 3 Ans

8g) -1 divided by 3

sol

$$-1 = (-1)(3) + 2$$

$\therefore$ Quotient = -1 Remainder = 2 Ans

h) 4 divided by 1

sol

$$4 = 4(1) + 0$$

$\therefore$ Quotient = 4 Remainder = 0 Ans

Q8 ii) $a = -111$ $m = 99$

sol

$$-111 = (-2)(99) + 87$$

$\therefore a \text{ div } m = -2$

$a \text{ mod } m = 87$

Ans

b) $a = -9999$ $m = 101$

sol

$$-9999 = -99(101) + 0$$

$\therefore a \text{ div } m = -99$

$a \text{ mod } m = 0$

Ans

c) $a = 10299$ $m = 999$

sol

$$10299 = 10(999) + 309$$

$\therefore a \text{ div } m = 10$

$a \text{ mod } m = 309$

Ans

d) $a = 123456 \quad m = 1001$

sol
 $123456 = 123(1001) + 333$

$\therefore a \text{ div } m = 123$

$a \text{ mod } m = 333$

Q8iii) $80 \text{ mod } 17 = 12$

as $12 \neq 5 \therefore 80$ is NOT congruent to $5 \text{ mod } 17$

b) $103 \text{ mod } 17 = 1$

as $1 \neq 5 \therefore 103$ is NOT congruent

c) $-29 \text{ mod } 17 = 5$

sol
 $-29 = (-2)(17) + 5$
↳ remainder

$\therefore -29$ is congruent

d) $-122 \text{ mod } 17 = 14$

as $14 \neq 5 \therefore -122$ is NOT congruent

Q9ai) 11, 15, 19

$\text{gcd}(11, 15)$

$15 = 1(11) + 4$

$11 = 2(4) + 3$

$4 = 1(3) + 1$

$3 = 3(1) + 0$

$\text{gcd}(11, 15) = 1$

$\text{gcd}(11, 19)$

$19 = 1(11) + 8$

$11 = 1(8) + 3$

$8 = 2(3) + 2$

$3 = 1(2) + 1$

$2 = 2(1) + 0$

$\text{gcd}(11, 19) = 1$

$\text{gcd}(15, 19)$

$19 = 1(15) + 4$

$15 = 3(4) + 3$

$4 = 1(3) + 1$

$3 = 3(1) + 0$

$\text{gcd}(15, 19) = 1$

As each pair gcd is equal to 1 it's a pairwise relative prime

9a ii) 14, 15, 21

$$\gcd(14, 15)$$

$$15 = 1(14) + 1$$

$$14 = 14(1) + 0$$

$$\gcd(14, 15) = 1$$

$$\gcd(14, 21)$$

$$21 = 1(14) + 7$$

$$14 = 2(7) + 0$$

$$\gcd(14, 21) = 7$$

$$\gcd(15, 21)$$

$$21 = 1(15) + 6$$

$$15 = 2(6) + 3$$

$$6 = 2(3) + 0$$

$$\gcd(15, 21) = 3$$

They are not pairwise relative prime as $(14, 21)$ has gcd 7 and $(15, 21)$ has gcd 3. and Not 1

aiii) 12, 17, 31, 37

$$\gcd(12, 17)$$

$$17 = 1(12) + 5$$

$$12 = 2(5) + 2$$

$$5 = 2(2) + 1$$

$$2 = 2(1) + 0$$

$$\gcd(12, 17) = 1$$

$$\gcd(12, 31)$$

$$31 = 2(12) + 7$$

$$12 = 1(7) + 4$$

$$7 = 1(4) + 3$$

$$4 = 1(3) + 1$$

$$3 = 3(1) + 0$$

$$\gcd(12, 31) = 1$$

$$\gcd(12, 37)$$

$$37 = 3(12) + 1$$

$$12 = 12(1) + 0$$

$$\gcd(12, 37) = 1$$

$$\gcd(17, 31)$$

$$31 = 1(17) + 14$$

$$17 = 1(14) + 3$$

$$14 = 4(3) + 2$$

$$3 = 1(2) + 1$$

$$2 = 2(1) + 0$$

$$\gcd(17, 31) = 1$$

$$\gcd(17, 37)$$

$$37 = 2(17) + 3$$

$$17 = 5(3) + 2$$

$$3 = 1(2) + 1$$

$$2 = 2(1) + 0$$

$$\gcd(17, 37) = 1$$

$$\gcd(31, 37)$$

$$37 = 1(31) + 6$$

$$31 = 5(6) + 1$$

$$6 = 1(6) + 0$$

$$\gcd(31, 37) = 1$$

As all pairs gcd is equal to 1 they are indeed pairwise relative prime.

Q9iv) 7, 8, 9, 11

$\gcd(7, 8)$ $8 = 1(7) + 1$ $7 = 7(1) + 0$ $\gcd(7, 8) = 1$	$\gcd(7, 9)$ $9 = 1(7) + 2$ $7 = 3(2) + 1$ $2 = 2(1) + 0$ $\gcd(7, 9) = 1$	$\gcd(7, 11)$ $11 = 1(7) + 4$ $7 = 1(4) + 3$ $4 = 1(3) + 1$ $3 = 3(1) + 0$ $\gcd(7, 11) = 1$	$\gcd(8, 9)$ $9 = 1(8) + 1$ $8 = 8(1) + 0$ $\gcd(8, 9) = 1$
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$\gcd(8, 11)$ $11 = 1(8) + 3$ $8 = 2(3) + 2$ $3 = 1(2) + 1$ $2 = 2(1) + 0$ $\gcd(8, 11) = 1$	$\gcd(9, 11)$ $11 = 1(9) + 2$ $9 = 4(2) + 1$ $2 = 2(1) + 0$ $\gcd(9, 11) = 1$
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As each pair's gcd equal to 1, therefore they are a pairwise relative prime.

Q9bi) $88 = 2^3 \cdot 11$

i) $126 = 2 \cdot 3^2 \cdot 7$

iii) $729 = 3^6$

iv) $1001 = 7 \cdot 11 \cdot 13$

v) $1111 = 11 \cdot 101$

vi) $909 = 3^2 \cdot 101$

working

rough work

$12 \overline{) 88}$ $2 \overline{) 8}$ $2 \overline{) 4}$ $2 \overline{) 2}$ $11 \times 2 \times 2 \times 2$ 11×2^3	$7 \overline{) 126}$ $2 \overline{) 18}$ $3 \overline{) 9}$ $3 \overline{) 3}$ $7 \times 2 \times 3 \times 3$ $7 \times 2 \times 3^2$	$3 \overline{) 729}$ $3 \overline{) 243}$ $3 \overline{) 81}$ $2 \overline{) 27}$ $3 \overline{) 9}$ $3 \overline{) 3}$ $3 \times 3 \times 3 \times 3 \times 3 \times 3$ 3^6
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actual

$2 \overline{) 88}$ $2 \overline{) 44}$ $2 \overline{) 22}$ $11 \overline{) 11}$ $2^3 \cdot 11$	$2 \overline{) 126}$ $3 \overline{) 63}$ $3 \overline{) 21}$ $7 \overline{) 7}$ $2 \times 3^2 \times 7$	$3 \overline{) 729}$ $3 \overline{) 243}$ $3 \overline{) 81}$ $3 \overline{) 27}$ $3 \overline{) 9}$ $3 \overline{) 3}$ 3^6
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$7 \overline{) 1001}$ $11 \overline{) 143}$ $13 \overline{) 13}$ $7 \cdot 11 \cdot 13$	$11 \overline{) 1111}$ $101 \overline{) 101}$ $11 \cdot 101$	$3 \overline{) 909}$ $3 \overline{) 303}$ $101 \overline{) 101}$ $3^2 \cdot 101$
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Q9c i) 11, 15, 105

$$\begin{array}{r} 11 \\ 11 \end{array} \quad \begin{array}{r} 3 \cdot 15 \\ 5 \cdot 3 \\ 1 \end{array} \quad \begin{array}{r} 3 \cdot 105 \\ 5 \cdot 21 \\ 7 \cdot 3 \\ 1 \end{array}$$

Prime Factors $\Rightarrow 11 = 11$

$$15 = 3 \cdot 5$$

$$105 = 3 \cdot 5 \cdot 7$$

$$\gcd(11, 15, 105) = 1$$

$$\text{lcm}(11, 15, 105) = 3 \cdot 5 \cdot 7 \cdot 11 = 1155$$

ii) 14, 15, 21

$$\begin{array}{r} 2 \cdot 14 \\ 7 \cdot 2 \\ 1 \end{array} \quad \begin{array}{r} 3 \cdot 21 \\ 7 \cdot 3 \\ 1 \end{array}$$

Prime factors: $14 = 2 \cdot 7$

$$15 = 3 \cdot 5$$

$$21 = 3 \cdot 7$$

$$\gcd(14, 15, 21) = 2^0 \cdot 3^0 \cdot 5^0 \cdot 7^0 = 1$$

$$\text{lcm}(14, 15, 21) = 2^1 \cdot 3^1 \cdot 5 \cdot 7 = 210$$

iii) 12, 72, 31, 32

Prime factors $\Rightarrow 12 = 2^2 \cdot 3$

$$72 = 2^3 \cdot 3^2$$

$$31 = 31$$

$$32 = 2^5$$

$$\gcd(12, 72, 31, 32) = 2^0 \cdot 3^0 \cdot 31^0 = 1$$

$$\text{lcm}(12, 72, 31, 32) = 2^{\max(2, 3, 0, 5)} \cdot 3^{\max(1, 2, 0, 0)} \cdot 31^{\max(0, 0, 1, 0)}$$

$$= 2^5 \cdot 3^2 \cdot 31$$

$$= 8928$$

$$\begin{array}{r} 2 \cdot 12 \\ 3 \cdot 4 \\ 1 \end{array} \quad \begin{array}{r} 2 \cdot 72 \\ 2 \cdot 36 \\ 2 \cdot 18 \\ 3 \cdot 9 \\ 3 \cdot 3 \\ 1 \end{array} \quad \begin{array}{r} 31 \cdot 31 \\ 1 \end{array} \quad \begin{array}{r} 2 \cdot 32 \\ 2 \cdot 16 \\ 2 \cdot 8 \\ 2 \cdot 4 \\ 2 \cdot 2 \\ 1 \end{array}$$

iv) 17, 18, 192, 1021

Prime factors $\Rightarrow 17 = 17$

$$18 = 2 \cdot 3^2$$

$$192 = 2^6 \cdot 3$$

$$1021 = 1021$$

$$\gcd(17, 18, 192, 1021) = 2^0 \cdot 3^0 \cdot 17^0 \cdot 1021^0 = 1$$

$$\text{lcm}(17, 18, 192, 1021) = 2^{\max(0, 1, 6, 0)} \cdot 3^{\max(0, 2, 1, 0)} \cdot 17^{\max(1, 0, 0, 0)} \cdot 1021^{\max(0, 0, 0, 1)}$$

$$= 2^6 \cdot 3^2 \cdot 17 \cdot 1021$$

$$= 9997632$$

$$\begin{array}{r} 17 \cdot 17 \\ 1 \end{array} \quad \begin{array}{r} 2 \cdot 18 \\ 3 \cdot 6 \\ 3 \cdot 3 \\ 1 \end{array} \quad \begin{array}{r} 2 \cdot 192 \\ 2 \cdot 96 \\ 2 \cdot 48 \\ 2 \cdot 24 \\ 2 \cdot 12 \\ 2 \cdot 6 \\ 3 \cdot 2 \\ 1 \end{array} \quad \begin{array}{r} 1021 \cdot 1021 \\ 1 \end{array}$$

Q10ai) GCD(144, 89)

$$144 = 1(89) + 55$$

$$89 = 1(55) + 34$$

$$55 = 1(34) + 21$$

$$34 = 1(21) + 13$$

$$21 = 1(13) + 8$$

$$13 = 1(8) + 5$$

$$8 = 1(5) + 3$$

$$5 = 1(3) + 2$$

$$3 = 1(2) + 1$$

$$2 = 2(1) + 0$$

Now finding linear combination

~~$$1(144) - 1(89) = 55$$~~

$$1(144) - 1(89) = 55$$

$$1(89) - 1(55) = 34 \rightarrow 1(55) - 1(34) = 21$$

$$1(34) - 1(21) = 13$$

$$1(21) - 1(13) = 8$$

$$1(13) - 1(8) = 5$$

$$1(8) - 1(5) = 3$$

$$1(5) - 1(3) = 2$$

$$1(3) - 1(2) = 1$$

~~$$1(2) - 2(1) = 0$$~~

Now substituting:

$$1(3) - 1(2) = 1$$

$$1(3) - 1(1(5) - 1(3)) = 1$$

$$2(3) - 1(5) = 1$$

$$2(1(8) - 1(5)) - 1(1(13) - 1(8)) = 1$$

$$2(8) - 2(1(13) - 1(8)) - 1(13) + 1(8) = 1$$

$$5(8) - 3(13) = 1$$

$$5(1(21) - 1(13)) - 3(13) = 1$$

$$5(21) - 8(13) = 1$$

~~$$5(21) - 8(13) = 1$$~~

$$5(1(55) - 1(34)) - 8(1(34) - 1(21)) = 1$$

$$5(55) - 5(34) - 8(34) + 8(21) = 1$$

$$5(55) - 13(34) + 8(55) - 1(34) = 21$$

$$5(1(144) - 1(89)) - 13(1(89) - 1(55)) + 8(55) - 8(34) = 21$$

$$5(144) - 5(89) - 13(89) + 13(1(144) - 1(89)) + 8(1(144) - 1(89)) - 8(34) = 21$$

$$26(144) - 39(89) - 8(34) = 1$$

$$26(144) - 39(89) - 8(1(89) - 1(55)) = 1$$

$$26(144) - 39(89) - 8(89) + 8(1(144) - 1(89)) = 1$$

$$34(144) - 55(89) = 1$$

10ai) GCD(1001, 100001)

$$100001 = 99(1001) + 902$$

$$1001 = 1(902) + 99$$

$$902 = 9(99) + 11$$

$$99 = 9(11) + 0$$

So GCD = 11

Substituting :-

$$1(902) - 9(99) = 11$$

$$1(1(100001) - 99(1001)) - 9(1(1001) - 1(902)) = 11$$

$$1(100001) - 99(1001) - 9(1001) + 9(1(100001) - 99(1001)) = 11$$

$$10(100001) - 999(1001) = 11 \quad \text{Ans}$$

Now linear combination

$$1(100001) - 99(1001) = 902$$

$$1(1001) - 1(902) = 99$$

$$1(902) - 9(99) = 11$$

bi) $55x \equiv 34 \pmod{89}$

finding inverse

$$89 = 1(55) + 34$$

$$55 = 1(34) + 21$$

$$34 = 1(21) + 13$$

$$21 = 1(13) + 8$$

$$13 = 1(8) + 5$$

$$8 = 1(5) + 3$$

$$5 = 1(3) + 2$$

$$3 = 1(2) + 1$$

$$2 = 2(1) + 0$$

Backwards

~~$$1(3) - 1(2) = 1$$~~

$$1(3) - 1(2) = 1$$

$$1(5) - 1(3) = 2$$

$$1(8) - 1(5) = 3$$

$$1(13) - 1(8) = 5$$

$$1(21) - 1(13) = 8$$

$$1(34) - 1(21) = 13$$

$$1(55) - 1(34) = 21$$

$$1(89) - 1(55) = 34$$

$$\gcd(55, 89) = 1$$

$\therefore$ Inverse exist

$$1(3) - 1(2) = 1$$

$$1(1(8) - 1(5)) - (1(5) - 1(3)) = 1$$

$$1(8) - 2(5) + 1(3) = 1$$

$$1(8) - 2(5) + 1(8) - 1(5) = 1$$

$$2(8) - 3(5) = 1$$

$$2(1(21) - 1(13)) - 3(1(13) - 1(8)) = 1$$

$$2(21) - 2(13) - 3(13) + 3(8) = 1$$

$$2(21) - 5(1(34) - 1(21)) + 3(1(21) - 1(13)) = 1$$

~~$$2(21) - 5(34) - 3(13) = 1$$~~

$$10(21) - 5(34) - 3(1(34) - 1(21)) = 1$$

$$-8(34) + 13(21) = 1$$

$$-8(1(89) - 1(55)) + 13(1(55) - 1(34)) = 1$$

$$-8(89) + 8(55) + 13(55) - 13(34) = 1$$

$$-8(89) + 21(55) - 13(1(89) - 1(55)) = 1$$

$$-21(89) + 34(55) = 1$$

$\therefore$ modular inverse of

55 modulo 89 is

$$55^{-1} \equiv 34 \pmod{89}$$

$$\begin{aligned}
 Q \quad & 2(21) - 5(34) - 5(21) - 3(21) + 3(13) = 1 \\
 & -6(21) - 5(34) + 3(1(34) - 1(21)) = 1 \\
 & -9(21) - 2(34) = 1 \\
 & -9(1(55) - 1(34)) - 2(1(89) - 2(55)) = 1
 \end{aligned}$$

Solve for x

$$55x \equiv 34 \pmod{89}$$

multiply both sides by $55^{-1} = 34$

$$55(55^{-1})x = 34 \times 34 \pmod{89}$$

$$x = 1156 \pmod{89}$$

$$x = 88 \pmod{89}$$

$$ii) \quad 89x = 2 \pmod{232}$$

$$\text{GCD}(89, 232)$$

$$232 = 2(89) + 54$$

$$89 = 1(54) + 35$$

$$54 = 1(35) + 19$$

$$35 = 1(19) + 16$$

$$19 = 1(16) + 3$$

$$16 = 5(3) + 1$$

$$3 = 3(1) + 0$$

$$\text{GCD} = 1$$

inverse exist

linear combination

$$1(232) - 2(89) = 54$$

$$1(89) - 1(54) = 35$$

$$1(54) - 1(35) = 19$$

$$1(35) - 1(19) = 16$$

$$1(19) - 1(16) = 3$$

$$1(16) - 5(3) = 1$$

Substituting

$$1(16) = 5(3) + 1$$

$$1(1(35) - 1(19)) - 5(1(19) - 1(35) - 1(19)) = 1$$

$$1(35) + 1(19) + 5(35) - 10(19) = 1$$

$$6(35) - 11(19) = 1$$

$$6(1(232) - 2(89)) - 11(1(54) - 1(35)) = 1$$

$$6(232) - 12(89) + 11(54) + 11(35) = 1$$

$$6(232) - 12(89) - 11(1(232) - 2(89)) + 11(1(89) - 1(1(232) - 2(89))) = 1$$

$$6(232) - 12(89) = 11(232) + 22(89) + 11(89) - 11(232) + 22(89) = 1$$

$$6(1(89) - 1(54)) - 11(1(54) - 1(35)) = 1$$

$$6(89) - 6(54) - 11(54) + 11(35) = 1$$

$$6(89) - 17(1(232) - 2(89)) + 11(1(89) - 1(54)) = 1$$

$$-28(89) - 17(232) + 11(89) - 11(54) = 1$$

$$-17(232) - 17(89) - 11(1(232) - 2(89)) = 1$$

$$-28(232)$$

$$1(16) - 5(3) = 1$$

$$1(1(35) - 1(19)) - 5(1(19) - 1(16)) = 1$$

$$1(35) - 1(19) - 5(19) + 5(16) = 1$$

$$1(35) - 6(1(54) - 1(35)) + 5(1(35) - 1(19)) = 1$$

$$1(35) - 6(54) + 6(35) + 5(35) - 5(1(54) - 1(35)) = 1$$

$$-11(54) + 17(35) = 1$$

$$-11(1(232) - 2(89)) + 17(1(89) - 1(54)) = 1$$

$$-11(232) + 22(89) + 17(89) - 17(54) = 1$$

$$-11(232) + 39(89) - 17(1(232) - 2(89)) = 1$$

$$-28(232) + 73(89) = 1$$

↳ inverse

$$89(73)x = 2(73) \pmod{232}$$

$$x = 146 \pmod{232}$$

(i) $a = 2, m = 17$

$$\text{GCD}(2, 17)$$

$$ax \equiv 1 \pmod{m}$$

$$17 = 8(2) + 1$$

$$2 = 2(1) + 0$$

so

$$2x \equiv 1 \pmod{17}$$

$$\text{GCD} = 1 \quad \text{inverse exist}$$

linear combination

$$1(17) - 8(2) = 1$$

$$\text{so } -8 + 17 = 9$$

$$2^{-1} \pmod{17}$$

~~multiply both side~~

$$\text{Hence } \bar{a} = 9$$

cii) $a = 34$ $m = 89$

$\text{GCD}(89, 34)$

$$89 = 2(34) + 21$$

$$34 = 1(21) + 13$$

$$21 = 1(13) + 8$$

$$13 = 1(8) + 5$$

$$8 = 1(5) + 3$$

$$5 = 1(3) + 2$$

$$3 = 1(2) + 1$$

$$2 = 2(1) + 0$$

$\text{GCD} = 1$ so inverse exist

$$1(3) - 1(2) = 1$$

$$1(5) - 1(3) = 2$$

$$1(8) - 1(5) = 3$$

$$1(13) - 1(8) = 5$$

$$1(21) - 1(13) = 8$$

$$1(34) - 1(21) = 13$$

$$1(89) - 2(34) = 21$$

substitute

$$1(3) - 1(2) = 1$$

$$1(1(8) - 1(5)) - 1(1(5) - 1(3)) = 1$$

$$1(8) - 2(5) + 1(3) = 1$$

$$1(1(21) - 1(13)) - 2(1(13) - 1(8)) + 1(1(8) - 1(5)) = 1$$

$$1(21) - 1(13) - 2(13) + 3(8) - 1(5) = 1$$

$$1(21) - 3(13) + 3(1(21) - 1(13)) - 1(1(13) - 1(8)) = 1$$

$$1(21) - 3(13) + 3(21) - 3(13) - 1(13) + 1(8) = 1$$

$$4(21) - 7(13) + 1(1(21) - 1(13)) = 1$$

$$5(21) - 8(13) = 1$$

$$5(1(89) - 2(34)) - 8(1(34) - 1(1(89) - 2(34))) = 1$$

$$5(89) - 10(34) - 8(34) + 8(89) - 16(34) = 1$$

$$13(89) - 34(34) = 1$$

Inverse = $-34 + 89$

$\bar{a} = 55$

ciii) $a = 144$ $m = 233$

$\text{GCD}(233, 144)$

$$233 = 1(144) + 89$$

$$144 = 1(89) + 55$$

$$89 = 1(55) + 34$$

$$55 = 1(34) + 21$$

$$34 = 1(21) + 13$$

$$21 = 1(13) + 8$$

$$13 = 1(8) + 5$$

$$8 = 1(5) + 3$$

$$5 = 1(3) + 2$$

$$3 = 1(2) + 1$$

$$2 = 2(1) + 0$$

gcd = 1 inverse exist

till 21 iam copying
cii sol as it's same

Backward

∴ same as cii

$$5(21) - 8(13) = 1$$

$$5(1(55) - 1(34)) - 8(1(34) - 1(21)) = 1$$

$$5(55) - 5(34) - 8(34) + 8(21) = 1$$

$$5(1(144) - 1(89)) - 13(1(89) - 1(55)) + 8(1(55) - 1(34)) = 1$$

$$5(144) - 5(89) - 13(89) + 13(55) + 8(55) - 8(1(89) - 1(55)) = 1$$

$$5(144) - 26(89) + 29(55) = 1$$

$$5(144) - 26(1(233) - 1(144)) + 29(1(144) - 1(89)) = 1$$

$$31(144) - 26(233) + 29(144) - 29(1(233) - 1(144)) = 1$$

$$60(144) + 29(144) - 26(233) - 29(233) = 1$$

$$89(144) - 55(233) = 1$$

Modular inverse of $144 \bmod 233$ is 89 Ans

civ) $a = 200, m = 1001$

$$\text{GCD}(1001, 200)$$

$$1001 = 5(200) + 1$$

$$200 = 200(1) + 0$$

$$\text{GCD} = 1 \text{ inverse exist}$$

Linear combination

$$1(1001) - 5(200) = 1$$

$$\therefore \text{Inverse} = -5 + 1001 \\ = 996 \text{ Ans}$$

Q11ai) Check Digit Value

$$X_{10} = 1(1) + 2(2) + 3(5) + 4(9) + 5(7) + 6(3) + 7(1) + 8(2) + 9(8) \bmod (11)$$

$$X_{10} = 204 \bmod 11$$

$$X_{10} = 6$$

ii) $7 \times 15 = 105$

$$105 \bmod 26 = 1$$

$$\begin{array}{r} 4 \\ 26 \overline{) 105} \\ \underline{104} \\ 1 \end{array}$$

$7 \times 15 \equiv 1 \pmod{26}$ hence shown proof

bi) $n = 11 \cdot 3 = 33$ $e = 3$

$\begin{matrix} p' & b \\ p & q \end{matrix}$

$n = 33$

going with

$a = 1, b = 2, c = 3$ and
 $e = 4, f = 5, g = 6$ so on

"exam" = $[5, 24, 1, 13]$

RSA $\Rightarrow C \equiv m^e \pmod{n}$

$C_1 = 5^3 \pmod{33} = 125 \pmod{33} = 26$

$C_2 = 24^3 \pmod{33} = 13824 \pmod{33} = 30$

$C_3 = 1^3 \pmod{33} = 1 \pmod{33} = 1$

$C_4 = 13^3 \pmod{33} = 2197 \pmod{33} = 19$

cipher text = $[26, 30, 1, 19]$ dec

ii) encrypted; "XYTU YFQ PNSL"

$f(p) = (p+5) \pmod{26}$ } encryption

Let $A = 0$

$X = 23$

$Y = 24$

$T = 19$

$U = 20$

$Y = 24$

$F = 5$

$Q = 16$

$P = 15$

$N = 13$

$S = 18$

$L = 11$

Decryption

$p = f(p) - 5 \pmod{26}$

$X \rightarrow 23 - 5 \pmod{26} = 18$ (S)

$Y \rightarrow 24 - 5 \pmod{26} = 19$ (T)

$T \rightarrow 19 - 5 \pmod{26} = 14$ (O)

$U \rightarrow 20 - 5 \pmod{26} = 15$ (P)

$Y \rightarrow 24 - 5 \pmod{26} = 19$ (T)

$F \rightarrow 5 - 5 \pmod{26} = 0$ (A)

$Q \rightarrow 16 - 5 \pmod{26} = 11$ (L)

$P \rightarrow 15 - 5 \pmod{26} = 10$ (K)

$N \rightarrow 13 - 5 \pmod{26} = 8$ (I)

$S \rightarrow 18 - 5 \pmod{26} = 13$ (N)

$L \rightarrow 11 - 5 \pmod{26} = 6$ (G)

decrypted text: STOP TALKING

(i) $X \equiv -1 \pmod{25}$

all x such that

$$\frac{x+1}{25} = c \rightarrow \text{eq 1}$$

or $25 \mid (x+1)$ same thing

from eq 1

$$x+1 = 25c$$

$$x = 25c - 1$$

$$c=0 \quad x=-1$$

$$c=1 \quad x=24$$

$$c=2 \quad x=49$$

$$c=3 \quad x=74$$

$$c=4 \quad x=99$$

$$c=-1 \quad x=-26$$

$$c=-2 \quad x=-51$$

$$c=-3 \quad x=-76$$

$$\therefore -76, -51, -26, -1, 24, 49, 74, 99$$

(ii) $5^{119} \pmod{59}$

According to Fermat little theorem

$$\therefore a^{p-1} \equiv 1 \pmod{p}$$

$$5^{58} \equiv 1 \pmod{59}$$

59 is prime

$$a=5$$

$$p-1 = 59-1 = 58$$

then,

$$5^{119} = (5^{58})^2 \cdot 5^3$$

$$(5^{58})^2 \cdot 5^3 \equiv x \pmod{59}$$

$$1 \cdot 5^3 \equiv x \pmod{59}$$

$$125 \equiv 7 \pmod{59}$$

(Q12ai) $x \equiv \pmod{5}$

$$x \equiv 2 \pmod{6}$$

$$x \equiv 3 \pmod{7}$$

$$M = m_1 \times m_2 \times m_3$$

$$M = 5 \times 6 \times 7$$

$$M = 210$$

$$\text{so, } x = M_1 M_1^{-1} a_1 + M_2 M_2^{-1} a_2 + M_3 M_3^{-1} a_3 \pmod{M}$$

$$M_1 = \frac{M}{m_1} = \frac{210}{5} = 42$$

$$M_2 = \frac{210}{6} = 35$$

$$M_3 = \frac{210}{7} = 30$$

for inverses

$$M_1 M_1^{-1} \equiv 1 \pmod{m_1}$$

$$42 M_1^{-1} \equiv 1 \pmod{5}$$

$$42 \cdot 3 \equiv 1 \pmod{5}$$

$$M_1^{-1} = 3$$

$$M_2 M_2^{-1} \equiv 1 \pmod{m_2}$$

$$35 M_2^{-1} \equiv 1 \pmod{6}$$

$$35 \cdot 5 \equiv 1 \pmod{6}$$

$$M_2^{-1} = 5$$

$$M_3 M_3^{-1} \equiv 1 \pmod{7}$$

$$30 \cdot 4 \equiv 1 \pmod{7}$$

$$M_3^{-1} = 4$$

$$X = 42 \cdot 3 \cdot 1 + 35 \cdot 2 \cdot 2 + 30 \cdot 4 \cdot 3 \pmod{210}$$

$$X = 836 \pmod{210}$$

$$X = 206 \pmod{210}$$

$$\text{ii)} \quad X \equiv 1 \pmod{2} \quad X \equiv 2 \pmod{3} \quad X \equiv 3 \pmod{5}$$

$$X \equiv 4 \pmod{11}$$

$$M = 2 \times 3 \times 5 \times 11 = 330$$

$$M_1 = \frac{M}{m_1} = 165 \quad ; \quad M_2 = 110 \quad ; \quad M_3 = 66 \quad ; \quad M_4 = 30$$

inverse

$$M_1 M_1^{-1} \equiv 1 \pmod{m_1}$$

$$165 M_1^{-1} \equiv 1 \pmod{2}$$

$$M_1^{-1} = 1$$

$$M_2 M_2^{-1} \equiv 1 \pmod{m_2}$$

$$110 M_2^{-1} \equiv 1 \pmod{3}$$

$$M_2^{-1} = 2$$

$$M_3 M_3^{-1} \equiv 1 \pmod{m_3}$$

$$66 M_3^{-1} \equiv 1 \pmod{5}$$

$$M_3^{-1} = 1$$

$$M_4 M_4^{-1} \equiv 1 \pmod{m_4}$$

$$30 M_4^{-1} \equiv 1 \pmod{11}$$

$$M_4^{-1} = 7$$

$$X = 165 \cdot 1 \cdot 1 + 110(2)(2) + 66(1)(3) + 30(7)(4)$$

$$X = 1643 \pmod{330}$$

$$X = 323 \pmod{330} \quad \text{Ans}$$

$$\text{b ii)} \quad X \equiv 3 \pmod{5}$$

$$X \equiv 3 \pmod{6}$$

$$X \equiv 1 \pmod{7}$$

$$X \equiv 0 \pmod{11}$$

$$M = 5 \times 6 \times 7 \times 11$$

$$M = 2310$$

$$M_1 = \frac{M}{m_1} = 462 \quad ; \quad M_2 = 385$$

$$M_3 = 330$$

$$M_4 = 210$$

$$\left. \begin{array}{l} M_1 \cdot M_1^{-1} \equiv 1 \pmod{m_1} \\ 462 \cdot M_1^{-1} \equiv 1 \pmod{5} \\ M_1^{-1} = 3 \end{array} \right\} \left. \begin{array}{l} M_2 \cdot M_2^{-1} \equiv 1 \pmod{m_2} \\ 385 \cdot M_2^{-1} \equiv 1 \pmod{6} \\ M_2^{-1} = 1 \end{array} \right\} \left. \begin{array}{l} M_3^{-1} \cdot 330 \equiv 1 \pmod{7} \\ M_3^{-1} = 1 \end{array} \right\}$$

$$X = 462(3)(3) + 385(1)(3) + 330(1)(1) + 0 \pmod{2310}$$

$$X = 5643 \pmod{2310}$$

$$X = 1023 \pmod{2310} \text{ ans}$$

$$c) X \equiv 3 \pmod{4}$$

$$X \equiv 5 \pmod{9}$$

$$X \equiv 8 \pmod{13}$$

$$M = 4 \times 9 \times 13 = 468$$

$$M_1 = \frac{468}{4} = 117$$

$$M_2 = 52$$

$$M_3 = 36$$

$$M_1 \cdot M_1^{-1} \equiv 1 \pmod{m_1}$$

$$117 \cdot M_1^{-1} \equiv 1 \pmod{4}$$

$$M_1^{-1} = 1$$

$$M_2 \cdot M_2^{-1} \equiv 1 \pmod{m_2}$$

$$52 \cdot M_2^{-1} \equiv 1 \pmod{9}$$

$$M_2^{-1} = 4$$

$$M_3 \cdot M_3^{-1} \equiv 1 \pmod{m_3}$$

$$36 \cdot M_3^{-1} \equiv 1 \pmod{13}$$

$$M_3^{-1} = 4$$

$$X = 117(1)(3) + 52(4)(5) + 36(4)(8) \pmod{468}$$

$$X = 2543 \pmod{468}$$

$$X = 203 \pmod{468} \text{ ans}$$

Q13ai) I LOVE DISCRETE

8 11 14 21 4 38 18 2 17 4 19 4

MATHEMATICS

12 0 19 7 4 12 0 19 8 2 18

a → 0

b → 1

c → 2

d → 3

⋮

z = 25

Ceasar Cipher = $(p+3) \pmod{26}$

11 14 17 24 7 6 11 21 5 20 7 22 7 15 3 22 10 7 15 3 22

11 5 21

Translated:- L O R Y H G L V F U H W H P D W K H P D W L F V

11a) STOP POLLUTION

18 19 14 15 15 14 11 11 20 19 8 14 13

$$CC = p + 4 \pmod{26}$$

$$CC = \underline{22 \ 23 \ 18 \ 19} \quad \underline{14 \ 18 \ 15 \ 15 \ 25 \ 24 \ 12 \ 18 \ 17}$$

Translate:- WXST ToppZYMSR

11b) $CC = (p + 24) \pmod{26}$

$$CC = \underline{13 \ 14 \ 9 \ 10} \quad \underline{10 \ 9 \ 6 \ 6 \ 15 \ 14 \ 3 \ 9 \ 8}$$

$$CC = \text{NOJK KJGG POPJI}$$

bi) PLG WZR DVVLJQPHQW

MID TWO ASSIGNMENT } Decrypted

ii) IDVW QXFHVXQLYHUVLWB

FAST NUCES UNIVERSITY } Decrypted

ci) CEBBOXNOB XYG

SURRENDER NOW } Decrypted

ii) LO WI PBSOXN

BE MY FRIEND } Decrypted

Q14ai) $h(k) = k \pmod{97}$

$$h(k) = 34567981 \pmod{97}$$

$$h(k) = 91 \pmod{97} \text{ Ans}$$

ii) $h(183211232) = 183211232 \pmod{97}$
 $= 57$

iii) $h(220195744) = 220195744 \pmod{97}$
 $= 21$

iv) $h(987255335) = \cancel{\text{mod } 97} \ 987255335 \pmod{97}$
 $= 5$

$$\text{bi)} \quad h(104578690) = 104578690 \bmod 101 \\ = 58$$

$$\text{ii)} \quad h(432222187) = 432222187 \bmod 101 \\ = 60$$

$$\text{iii)} \quad h(372201919) = 372201919 \bmod 101 \\ = 52$$

$$\text{iv)} \quad h(501338753) = 501338753 \bmod 101 \\ = 3$$

$$\text{ci)} \quad X_{n+1} = (4X_n + 1) \bmod 7 \quad x_0 = 3$$

$$\underline{\text{sol}} \quad x_0 = 3$$

$$x_1 = 4(3) + 1 \bmod 7 = 13 \bmod 7 = 6$$

$$x_2 = 4(6) + 1 \bmod 7 = 25 \bmod 7 = 4$$

$$x_3 = 4(4) + 1 \bmod 7 = 17 \bmod 7 = 3$$

$$x_4 = 4(3) + 1 \bmod 7 = 13 \bmod 7 = 6$$

$$x_5 = 4(6) + 1 \bmod 7 = 25 \bmod 7 = 4$$

$$\text{ii)} \quad X_{n+1} = 7X_n + 2 \bmod 9$$

$$\underline{\text{sol}} \quad x_0 = 5$$

$$x_1 = 7(5) + 2 \bmod 9 = 37 \bmod 9 = 1$$

$$x_2 = 7(1) + 2 \bmod 9 = 9 \bmod 9 = 0$$

$$x_3 = 7(0) + 2 \bmod 9 = 2 \bmod 9 = 2$$

$$x_4 = 7(2) + 2 \bmod 9 = 16 \bmod 9 = 7$$

$$x_5 = 7(7) + 2 \bmod 9 = 51 \bmod 9 = 6$$

$$x_6 = 7(6) + 2 \bmod 9 = 44 \bmod 9 = 8$$

$$\text{Q15ai)} \quad 3(7) + 3 + 3(2) + 3 + 3(2) + 1 + 3(8) + 4 + 3(4) + 3 + 3(4) + x_{12} \equiv \\ 0 \bmod 10$$

$$95 + x_{12} \equiv 0 \bmod 10$$

$$\text{Check digit } (x_{12}) = 5 \quad \text{Ans}$$

$$ii) 3(6) + 3 + 3(6) + 2 + 3(3) + 9 + 3(9) + 1 + 3(3) + 4 + 3(6) + X_{12} \\ = 0 \pmod{10}$$

$$\overset{118}{126} + X_{12} = 0 \pmod{10}$$

$$\text{check digit } (X_{12}) = 2$$

$$bi) 3(0) + 3 + 3(6) + 0 + 3(0) + 0 + 3(2) + 9 + 3(1) + 4 + 3(5) + 2 \equiv 0 \pmod{10}$$

$$60 \equiv 0 \pmod{10} \quad \text{Valid UPC code}$$

$$ii) 3(0) + 1 + 3(2) + 3 + 3(4) + 5 + 3(6) + 7 + 3(8) + 9 + 3(0) + 3 \equiv 0 \pmod{10}$$

$$88 \equiv 0 \pmod{10} \quad \text{NOT valid}$$

$$ci) X_{10} = 1(0) + 2(0) + 3(7) + 4(1) + 5(1) + 6(9) + 7(8) + 8(8) + 9(1) \pmod{11}$$

$$X_{10} = 213 \pmod{11}$$

$$X_{10} = 4$$

$$X_{10} = \sum_{i=1}^9 i x_i \pmod{11}$$

$$ii) \sum_{i=1}^{10} i x_i \equiv 0 \pmod{11}$$

$$\Rightarrow 1(0) + 2(3) + 3(2) + 4(1) + 5(5) + 6(0) + 7(0) + 8Q + 9(1) + 10(8)$$

$$\equiv 0 \pmod{11}$$

$$130 + 8Q \equiv 0 \pmod{11}$$

$$\text{so } 130 = 9 \pmod{11}$$

$$\begin{array}{r} 11 \overline{) 130} \\ \underline{121} \\ 9 \end{array}$$

$$9 + 8Q \equiv 0 \pmod{11}$$

$$8Q \equiv -9 \pmod{11}$$

$$8Q \equiv 2 \pmod{11}$$

$$8(7) \equiv 1 \pmod{11}$$

$$Q \equiv$$

Multiply both side by 7

$$Q = 2 \cdot 7 = 14 = 3 \pmod{11}$$

$$Q = 3$$

$$\text{GCD}(11, 9)$$

$$11 = 1(9) + 2$$

$$9 = 3(2) + 3$$

$$1(11) - 1(9) = 2$$

$$-9 + 11 = 2$$

$$11 = 5(2) + 1$$

$$2 = 7(1) + 0$$

$$11 - 5(2) = 1$$

$$\textcircled{16a}) \quad n = 43 (59)$$

$$e = 13$$

$$\text{key} = (e, n)$$

"ATTACK"

"EXAMINATION"

$$C = M^e \bmod n$$

$$A \rightarrow C_1 = 0^{13} \bmod 2537 = 0$$

$$T \rightarrow C_2 = 19^{13} \bmod 2537 = 2299$$

$$C \rightarrow C_3 = 2^{13} \bmod 2537 = \cancel{1021} 8581$$

$$K \rightarrow C_4 = 10^{13} \bmod 2537 = 0$$

$$E \rightarrow C_5 = 4^{13} \bmod 2537 = 140$$

$$X \rightarrow C_6 = 23^{13} \bmod 2537 = 2475$$

$$M \rightarrow C_7 = 12^{13} \bmod 2537 = 2497$$

$$I \rightarrow C_8 = 8^{13} \bmod 2537 = 156$$

$$N \rightarrow C_9 = 13^{13} \bmod 2537 = 2038$$

$$O \rightarrow C_{10} = 14^{13} \bmod 2537 = 1616$$

Method 2

$$C(AT) = 019^{13} \bmod 2537$$

$$C(TA) = 140^{13} \bmod 2537$$

$$C(K) = 0210^{13} \bmod 2537$$

$$C(EX) = 0423^{13} \bmod 2537$$

$$C(EM) = 0012^{13} \bmod 2537$$

$$C(IN) = 0813 \bmod 2537$$

$$C(AT) = 0019^{13} \bmod 2537$$

$$C(IO) = 0814^{13} \bmod 2537$$

$$C(N) = 13^{13} \bmod 2537$$

b) ASSIGNMENT SEMESTER

$$n = 13(19) = 247$$

$$e = 47$$

$$C = M^{47} \bmod 247$$

$$A \rightarrow 0^{47} \bmod 247 = 0$$

$$S \rightarrow 18^{47} \bmod 247 = 151$$

$$I \rightarrow 8^{47} \bmod 247 = 31$$

$$G \rightarrow 6^{47} \bmod 247 = 245$$

$$N \rightarrow 13^{47} \bmod 247 = 78$$

$$M \rightarrow 12^{47} \bmod 247 = 103$$

$$E \rightarrow 4^{47} \bmod 247 = 244$$

$$T \rightarrow 9^{47} \bmod 247 = 76$$

$$R \rightarrow 17^{47} \bmod 247 = 23$$

c i) $5^{2003} \bmod 71$
 $5^{1918} \cdot 5^5 \bmod 7$
 $1 \cdot 5^5 \bmod 7$
 $5^5 \equiv 3 \bmod 7$

iii) $5^{2003} \bmod 13$
 $(5^{12})^{166} \cdot 5^{11} \bmod 13$
 $5^{11} \equiv 5 \bmod 13$

c ii) $5^{2003} \bmod 11$
 $(5^{10})^{200} \cdot 5^3 \bmod 11$
 $5^3 = 4 \bmod 11$

v) $5^{2003} \bmod 17$
 $(5^{10})^{125} \cdot 5^3 \bmod 17$
 $5^3 = 6 \bmod 17$

Q17a) $X = 2 \bmod 7$
 $X = 3 \bmod 17$
 $X = 5 \bmod 19$

$$M = 7 \times 17 \times 19 = 2261$$

$$M_1 = \frac{2261}{7} = 323 \quad ; \quad M_2 = \frac{2261}{17} = 133 \quad ; \quad M_3 = \frac{2261}{19} = 119$$

inverses

$$M_1 \cdot M_1^{-1} \equiv 1 \bmod m_1$$

$$323 \cdot M_1^{-1} \equiv 1 \bmod 7$$

$$M_1^{-1} = 1$$

$$M_3^{-1} \cdot 119 \equiv 1 \bmod 19$$

$$M_3^{-1} = 4$$

$$\left. \begin{array}{l} M_2^{-1} M_2 \equiv 1 \bmod m_2 \\ M_2 (133) \equiv 1 \bmod 17 \\ M_2^{-1} = 11 \end{array} \right\} \text{Euclidean}$$

$$\begin{array}{l} 133 = 7(17) + 14 \\ 1 = 3 - (14 - 4(3)) \quad (17 = 1(14) + 3) \\ 1 = -14 + 5(3) \quad 14 = 4(3) + 2 \\ 1 = -14 + 5(17 - 14) \quad 3 = 1(2) + 1 \\ 1 = 5(17) - 6(133 - 7(17)) \\ 1 = -6(133) + 47(17) \end{array}$$

$$X = 323(1)(2) + 133(11)(3) + 119(4)(5) \bmod 2261$$

$$X = 7415 \bmod 2261$$

$$X = 632 \bmod 2261$$

b) "SIX HUNDRED AND THIRTY TWO"
VLA KXQGUHG DQG WKLUNB WZR

$$c) X^{302} \bmod 7$$

$$X = \text{sum of digit of } 632$$

$$X = 6 + 3 + 2$$

$$X = 11$$

$$\therefore 11^{302} \bmod 7$$

Fermat theorem

$$11^{p-1} \equiv 1 \bmod 7$$

$$(11^6)^{50} \cdot 11^2 \bmod 7$$

$$1 \cdot 11^2 \bmod 7$$

$$121 \equiv 2 \bmod 7$$

Q18a) 27 floor, 37 office per floor

$$\begin{aligned} \text{Total office} &= \text{Floor} \times \text{office per floor} \\ &= 27 \times 37 \\ &= 999 \end{aligned}$$

$$b) 12C \times 2T \times 3S$$

$$= 72 \text{ types}$$

12 colors
2 version
3 size

$$c) \underbrace{26 \times 26 \times 26}$$

$$= 17576$$

$$Q19a) {}^{13}C_5 = \frac{13!}{5!(8!)} = \frac{13 \times 12 \times 11 \times 10 \times 9 \times 8!}{5 \times 4 \times 3 \times 2 \times 8!} = 1287$$

$$b) 75 \times 74 \times 73 \times 72 \times 71 = \cancel{205476450} 2071126800$$

$$c) {}^{27}C_4 = \frac{27!}{4!(23)!} = \frac{27 \times 26 \times 25 \times 24 \times 23!}{4 \times 3 \times 2 \times 23!} = 17550$$

Q20a) $16^{10} + 16^{26} + 16^{58}$ combination

$$b) 26^4 - 25^4 = 456976 - 390625 = 66351$$

↓ ↓
Total — without X

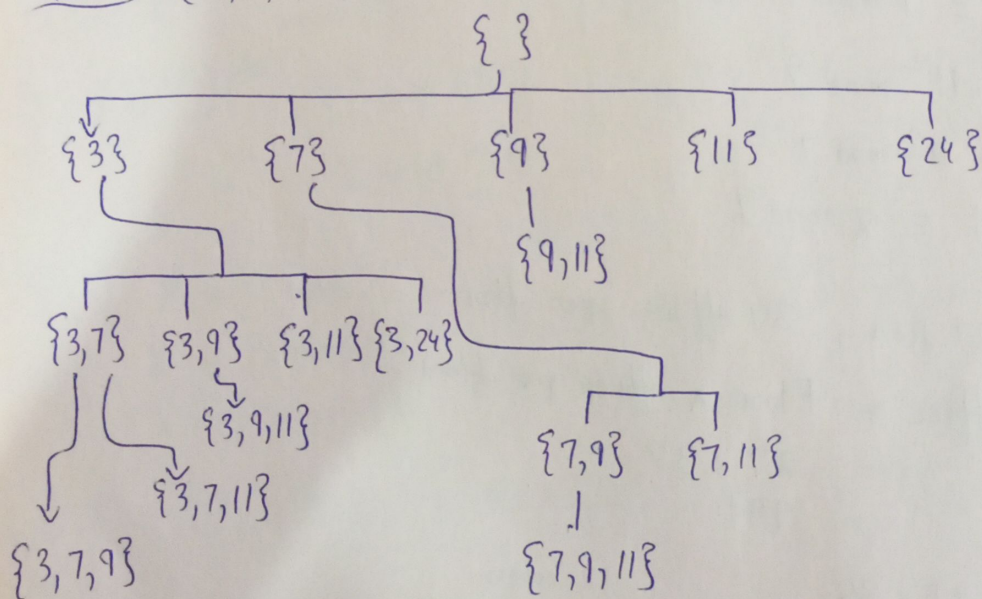
c) ~~$26 \times 25 \times 24$~~ $P(26, 3) = 26 \times 25 \times 24 = 15600$

Q21a) 2^m

b) $5! = 120$

c) $15 \times 48 \times 24 \times 34 \times 28 \times 28 = 460615680$

Q22a) $\{3, 7, 9, 11, 24\}$



b) Total $\Rightarrow 2^4 = 16$

with two consecutive 1's

$$\begin{array}{|c|c|c|} \hline 2 & 2 & 2 \\ \hline \end{array} = 2^3 = 8$$

w/o consecutiv = $16 - 8 = 8$

0000, 0001, 0010, 0100, 1000, 0101, 1010, 1001 } qty = 8

c) if C is secretary

President = $\{B, D\}$

Treasurer = $\{A, B, D\}$

$P = 2$

$T = 3 - 1 = 2$

so $2 \times 2 = 4$

If D is secretary

$$P = 2$$

$$T = 2$$

$$2 \times 2 = 4$$

$$\text{Total ways} = 4 + 4 = 8 \text{ Ans}$$

$$\text{Q23a)} \quad {}^8C_3 = \frac{8!}{3!5!} = \frac{8 \times 7 \times 6 \times 5!}{3 \times 2 \times 5!} = 56$$

$$\text{b)} \quad {}^{12}C_6 = \frac{12!}{6!6!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6!}{6 \times 5 \times 4 \times 3 \times 2 \times 6!} = 924$$

$$\text{c)} \quad {}^9C_5 = \frac{9!}{5!4!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4!}{5! \times 4 \times 3 \times 2} = 126$$

$$\text{Q24a)} \quad {}^{20}P_5 = \frac{20!}{15!} = \frac{20 \times 19 \times 18 \times 17 \times 16 \times 15!}{15!} = 1860480$$

$$\text{b)} \quad {}^{16}P_4 = \frac{16!}{12!} = \frac{16 \times 15 \times 14 \times 13 \times 12!}{12!} = 43680$$

$$\text{c)} \quad {}^{15}P_2 = \frac{15!}{13!} = \frac{15 \times 14 \times 13!}{13!} = 210$$

$$\text{Q25a)} \quad \text{meat} \rightarrow {}^5C_1 = 5$$

$$\text{cheese} \rightarrow {}^4C_1 = 4$$

$$\text{bread} \rightarrow {}^3C_2 = 3$$

$$\text{condiments} \rightarrow {}^6C_3 = 20$$

$$\boxed{\text{Total ways} = 5 \times 3 \times 4 \times 20 = 1200 \text{ Ans}}$$

$$\text{b)} \quad {}^{12}C_5 = \frac{12!}{5!7!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7!}{5 \times 4 \times 3 \times 2 \times 7!} = 792 \text{ Ans}$$

$$\text{ci)} \quad \text{with repetition: } \frac{1}{4} \underline{10} \underline{10} \underline{10} \underline{10} = 10^4 = 10000$$

$$\text{w/o repetition: } \underline{1} \underline{9} \underline{8} \underline{7} \underline{6} = 3024$$

ii) with repetition

$$\text{begin with 3: } \underline{1} \underline{10} \underline{10} \underline{10} \underline{10} = 10000$$

$$\text{begin with 1: } \underline{1} \underline{10} \underline{10} \underline{10} \underline{10} = 10000$$

$$\text{Total} = 10000 + 10000 = 20000$$

w/o repetition

$$\text{begin w/3} = \underline{1} \underline{9} \underline{8} \underline{7} \underline{6} = 3024$$

$$\text{begin w/1} = \underline{1} \underline{9} \underline{8} \underline{7} \underline{6} = 3024$$

$$\text{Total} = 3024 + 3024 = 6048$$

Q26a) begin with 000: $\underline{1} \underline{1} \underline{1} \mid \underline{2} \underline{2} \underline{2} \underline{2} \underline{2} \mid \underline{2} \underline{2} = 128$
end with 00: $\underline{2} \underline{2} \underline{2} \mid \underline{2} \underline{2} \underline{2} \underline{2} \underline{2} \mid \underline{1} \underline{1} = 256$

$$\text{Total} = \text{begin w/000} + \text{end with 00} - \text{common}$$
$$= 128 + 256 - 32 = 352 \text{ Ans}$$

b) begin with 0: $\underline{1} \mid \underline{2} \underline{2} \mid \underline{2} \underline{2} = 2^4 = 16$
end with 11: $\underline{2} \mid \underline{2} \underline{2} \mid \underline{1} \underline{1} = 2^3 = 8$

$$\text{Total} = 16 + 8 - 4 = 20$$

c) 2 letters, 4 digits

$$\text{with repetition} = \underline{26} \underline{26} \underline{10} \underline{10} \underline{10} \underline{10}$$
$$= 6760000 \text{ Ans}$$

$$\text{w/o repetition: } \underline{26} \underline{25} \underline{10} \underline{9} \underline{8} \underline{7}$$
$$= 3276000 \text{ Ans}$$

ii) 3 odd digits, 2 vowels

$$\text{with rep: } \underline{5} \times \underline{5} \times \underline{5} \times \underline{5} \times \underline{5} = 5^5 = 3125 \text{ Ans}$$

$$\text{without rep: } 5 \times 4 \times 3 \times 5 \times 4 = 1200 \text{ Ans}$$

odd digit 1, 3, 5, 7, 9 qty = 5 vowel a, e, i, o, u qty = 5

Q27a) Pigeon hole principle :- $\left\lceil \frac{\text{total student}}{\text{total letter}} \right\rceil$

$$\left\lceil \frac{30}{26} \right\rceil = 2$$

b) Pigeon hole = $\left\lceil \frac{\text{total population}}{\text{max hair}} \right\rceil = \frac{8008278}{1000001} = 9$

c) Pigeonhole = $\left\lceil \frac{\text{classes}}{\text{slots}} \right\rceil = \left\lceil \frac{67}{38} \right\rceil = 18$

Q28ai) coefficient x^5 in $(1+x)^{11}$

$r=5 \quad n=11 \quad x^{n-r} y^r$

${}^{11}C_5 (1)^{11-5} (x)^5$

$462(1)x^5$

462 coefficient

ii)

$a^7 b^{17}$ in $(2a-b)^{24}$

$r=17 \quad n=24$

${}^{24}C_{17} (2a)^{24-17} (-b)^{17}$

$346104 (2a)^7 (-b)^{17}$

$-346104 (128) (a^7 b^{17})$

$-44301312 a^7 b^{17}$

$\therefore -44301312$ coefficient

bi)

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 2 & & 1 \\
 & 1 & 3 & & 3 & & 1 \\
 1 & 4 & 6 & & 4 & & 1 \\
 1 & 5 & 10 & & 10 & & 5 & 1
 \end{array}$$

ii) $(3x-2)^4$

$1(3x)^4(-2)^0 + 4(3x)^3(-2)^1 + 6(3x)^2(-2)^2 + 4(3x)(-2)^3 + 1(3x)^0(-2)^4$

$1(3x)^0(-2)^4$

$= 3^4 x^4 + 4(-2)(3^3 x^3) + 6(4)(9x^2) - 8(4)(3x) + 16(1)$

$= 81x^4 - 216x^3 + 216x^2 - 96x + 16$

$$c) (2x+y)^6$$

$$= {}^6C_0 (2x)^{6-0} (y)^0 + {}^6C_1 (2x)^5 (y) + {}^6C_2 (2x)^4 (y^2) + {}^6C_3 (2x)^3 y^3 + {}^6C_4 (2x)^2 y^4 + {}^6C_5 (2x) y^5 + {}^6C_6 (2x)^0 y^6$$

$$= 1(2x)^6 + 6(2x^5y) + 15(16x^4y^2) + 20(8x^3y^3) + 15(4x^2y^4) + 6(2xy^5) + y^6$$

$$= 64x^6 + 192x^5y + 240x^4y^2 + 160x^3y^3 + 60x^2y^4 + 12xy^5 + y^6$$

$$ii) (3a-2b)^7$$

$$= {}^7C_0 (3a)^7 (-2b)^0 + {}^7C_1 (3a)^6 (-2b) + {}^7C_2 (3a)^5 (-2b)^2 + {}^7C_3 (3a)^4 (-2b)^3 + {}^7C_4 (3a)^3 (-2b)^4 + {}^7C_5 (3a)^2 (-2b)^5 + {}^7C_6 (3a) (-2b)^6 + {}^7C_7 (3a)^0 (-2b)^7$$

$$= 2187a^7 - 10206a^6b + 20412a^5b^2 - 22680a^4b^3 + 15120a^3b^4 - 6048a^2b^5 + 672ab^6 - 128b^7$$

$$Q29a) \text{ Men} = 16 \quad \text{Women} = 20 \quad \text{Total} = 36$$

$$\therefore 36!$$

$$ii) 7 \text{ students in a row} = {}^{36}P_7$$

$$iii) \text{ all student in the row women on left, men on right} = 20! \times 16!$$

$$b) \text{ Choices for pitches} \Rightarrow \{5, 6, 7, 8, 9\} \Rightarrow \text{qty} = 5$$

$$\& \text{ player arrangements} \rightarrow 8!$$

$$\text{Total} = 5 \times 8! = 201600 \text{ ans}$$

$$c) {}^{12}C_0 + {}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3$$

$$1 + 12 + 66 + 220$$

$$= 299 \text{ pizza ans}$$

$$\begin{aligned} \text{Q30a)} \quad & {}^{15}C_3 \times {}^{12}C_3 \times {}^7C_3 \times {}^5C_3 \times {}^9C_3 \times {}^{10}C_3 \\ &= 455 \times 220 \times 35 \times 10 \times 84 \times 120 \\ &= 353\,152\,800\,000 \text{ Ans} \end{aligned}$$

$$\begin{aligned} \text{b)} \quad & \frac{\text{total student} = 97}{\text{total grades} = 10} \quad \text{pigeonhole} \Rightarrow \left\lceil \frac{97}{10} \right\rceil = 10 \end{aligned}$$

$\therefore$ At least 10 students will get the same grade.

$$\begin{aligned} \text{c)} \quad & \text{When wicket keeper is captain} \rightarrow 2(2)(13) = 52 \\ & \text{When " " not captain} \rightarrow 13(12)(2) = 312 \end{aligned}$$

$$\text{Total} = 52 + 312 = 364 \quad \} \text{ without repetition}$$

With repetition

$$2 \times 15 \times 14 = 420 \text{ ways}$$

Two options $\swarrow \searrow$ anyone except chosen captain
anyone from squad

Q31a) Combination

- 1) In lottery matching
- 2) Forming cricket team

b) Permutation

- 1) Generate license plate
- 2) Arranging seatings

c) Binomial Theorem

- 1) Probability and statistics
- 2) Finance - modeling compound interest or option pricing in financial markets

d) Proof methods:

- 1) verification of software correction
- 2) prove security protocols

e) Mathematical Induction

- 1) For proving the correctness of recursive algorithm
- 2) Proving formulas